

Component 4 :

The primary objective of Component 4 is **not** to extract textual information from images, but to **associate price levels with non-native visual elements added to the chart** (drawings, annotations, user-added indicators).

Within the scope of the MVP, we focus exclusively on **charts (TradingView-like)**.

What we consider “key information” in an image

1) Associating prices with user-added graphical elements (most important)

Examples include:

- Arrows (up / down)
- Rectangles / boxes (support / resistance zones)
- Trendlines
- Channels
- Extensions / retracements (Fibonacci or manually drawn)
- Hand-drawn schemas

For each element, we want to be able to answer:

- What is the starting price?
- What is the ending price?
- Or, in the case of a zone: what are the lower and upper bounds?

Beyond extracting price levels associated with visual elements, Component 4 must also be able to **understand the expected price dynamics**, as expressed visually by the user on the chart.

The goal is not only to identify price levels, but also to **reconstruct the expected price path**, including the order and direction of the scenario anticipated by the user.

Is the movement:

- Direct (price moves from the current area)
- Conditional (a final lower low / higher high is expected before the move)

Does the scenario imply:

- A sweep / deviation before the move
- Immediate rejection
- Break & continuation
- Pullback then continuation

2) Textual information (secondary but useful)

- Ticker / asset (BTC, ETH, SOL, etc.)
- Timeframe
- Indicator name (if any)

Example 1

<https://x.com/IncomeSharks/status/1983240977218449521?s=20>

Asset: LAR (Lithium Argentina AG)

Timeframe: 1D

Visual elements:

- Element 1: an ascending trendline supporting higher lows
- Element 2: a box representing a key price zone between 3.55 and 4.00

Using the accompanying text, we understand that the user is considering adding to the position if price retests the trendline within this zone.

The trendline represents the **precise objective**, while the box represents a **broader acceptable zone**.

For this reason, we aim to determine the **price of the trendline at the time of the candle**.

Expected output:

- **Low Time Frame:** price may retrace into the 4.00–3.55 zone before resuming upward, with ideal support on the trendline around 3.80–3.85
- **High Time Frame:** bullish structure remains intact as long as price stays above the trendline

Remark: although the text suggests that the trendline is the most important level, the image clearly shows that price could dip below the trendline into the green box, and that this box actually represents the continuation zone that must hold for the bullish scenario to remain valid.

Overall conviction: medium

What we need to confirm in this example:

1. Bullish continuation on the medium timeframe
2. Was the buy zone reached?

This reflects a high level of control between corrective and continuation phases.

Process idea: draw vertical and horizontal lines to associate graphical objects with price levels as precisely as possible, using correct time alignment based on the relevant candles.



Example 2

<https://x.com/CryptoMillionYT/status/2002739391018488300?s=20>

Asset: BTC

Timeframe: H1

Visual elements: multiple lines, each to be associated with a price. Ideally, understanding the tag (EQL, EQH, FVG) would help distinguish targets from supports/resistances, but this is not strictly required for a good result. There is also a green zone linked to one of the lines, between 86,750 and 85,900. Finally, there is a hand-drawn path representing the expected near-term move.

The scenario suggests a move below the third line around 86,700, followed by a rebound and a new high.

The green zone is the expected bounce area, even though no clear invalidation is defined if price breaks through it.

For exits and targets, EQH levels are considered objectives to be reached.

There is also a green zone between 90,700 and 92,000, which appears to be the main target zone, even though the line itself extends higher.

What we need to confirm in this example:

- Did price reach the expected zone?
- If yes, how did it react afterward?
- Did it reach the projected targets?
- If not, was the final expected direction still correct?

Post-analysis conclusion:

Price did not reach the expected bounce zone and instead moved directly toward the target zone. The final direction was correct, but the immediate direction was wrong, making the entry impossible.

From the text, we also understand that the user was considering a lower entry, meaning they were significantly wrong on the short-term direction.

On a higher timeframe, the trade looks much better: price first moved up, then returned to the expected zone several days later, which became the true bottom before a strong multi-day rally. However, since the trade was framed on H1, the analysis is not valid.

Example 3

<https://x.com/TraderAAG/status/1998225678035210409?s=20>

Asset: BCH

Timeframe: H1

Visual elements: multiple lines and two overlapping blue zones (FVG + OB) around 561–562. There is also an arrow showing a downward move toward 559, followed by a rally toward the highs above the lines, around 607–608.

What we need to confirm:

- Did price reach the indicated zone?
- Did it bounce from that zone?
- Did it reach the expected targets?

Post-analysis conclusion:

Price did reach the zone and bounced, but failed to reach the targets and then continued lower, breaking below the zone and making a new low.

The analysis is therefore incorrect: the zone was interesting for a bounce, but the direction was wrong. This could have worked as a corrective move within a bearish trend, but not as a bullish continuation bottom.

Example 4

<https://x.com/KillaXBT/status/1922185937666109504?s=20>

Asset: BTC

Timeframe: not specified, but based on price levels it is clearly a very high timeframe (weekly)

Visual elements: manually drawn chart with price zones.

What we want to confirm:

The higher the timeframe, the less precise we expect the analysis to be. Here, the goal is primarily to understand direction and relative positioning of price zones.

The scenario describes:

- A bullish market up to a top above 120
- A trend change into a bearish market, with a correction to around 90, remaining above the last low of the previous bullish leg, followed by accumulation
- A further drop toward 70k, making a lower low than the previous bullish phase, followed by accumulation
- A third drop making a lower low than the previous correction

Post-analysis conclusion:

Bullish and bearish phases are broadly correct, even if sub-movements lack precision. The long-term direction, especially the bullish top, is very accurate.

The bearish move was slightly underestimated, with price reaching lower correction levels, but still respecting the overall logic: first correction above previous bullish lows, second below, and a final one below the prior low.

The following post completes this view and indicates a bottom for the last bearish phase around 45k, with a bearish range between 80k and 65k, followed by a bottoming range before returning to the current range (65k–80k):

<https://x.com/KillaXBT/status/2020503543095124381?s=20>

Example 5

<https://x.com/captaincrypto21/status/2016834058320216477?s=20>

Asset: ETH

Timeframe: H1

Visual elements: many indicators, lines, prices, and colors, but the key element is the hand-drawn curve.

The user expects a move back to 2,850 before making a new high.

Post-analysis conclusion:

The analysis is entirely wrong, as price continued to drop. However, the text shows that the user framed this as a possibility rather than a strong conviction.

Example 6

https://x.com/honey_xbt/status/1975197891737334193?s=20

Asset: TOTAL3 (an index, not a single asset)

Timeframe: 1W

Visual element: an arrow that is not coherent with the chart and should therefore be ignored.

The goal is simply to follow the index price, which is expected to rise in line with the user's "extremely bullish" view.

STRATEGY

- Price axis is always on the right
- Map pixel positions of each element (top and bottom)
- Use OpenCV (Canny + Hough) to identify pixels of the latest candles and graphical elements
- Project elements vertically (90°) onto the price axis
- Apply OCR only on the price axis to identify price values and their pixel positions
- Compute the price associated with each element pixel using proportional interpolation based on OCR-anchored values